

Configuration

Configure the neural network and the image source.

Select Model Weight

deepglobe_finetuned_model.pth

Image Source

- ☐ Fetch Random Image (Matches Model)
- ☒ Upload Custom Image

Upload Image

 3594_sat.jpg
383.5KB

×

+

Resilience Settings

Attack Simulation Type

- ☒ Targeted (Choke Points)
- ☐ Random (Chance)

Nodes to Remove (%) [Random Attack Only]

15

Geospatial Network Resilience Dashboard

Analyze satellite imagery, extract topology-preserving road networks, and simulate cascading infrastructure failures.

Start Analysis Pipeline

Terminal Execution Logs

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NETWORK RESILIENCE & DISASTER SIMULATION (GRAPH THEORY)

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[*] Automatically generating a new Multi-Spectral SpaceNet Mask...

===== CROSS-DATASET GENERALIZATION TEST =====

[*] Testing the model on unseen satellite data!

[*] Target Device: cpu

[*] Loading Brain Data from deepglobe_finetuned_model.pth...

[*] Loading custom image from temp_uploaded_img.jpg...

[*] Processing Image #custom

[*] Running Neural Network Prediction...

[*] Saved RGB Image to resilience/spacenet_generalize_img.jpg

[*] Saved Prediction Mask to resilience/spacenet_generalize_mask.png

[*] Loading satellite image crop: resilience/spacenet_generalize_img.jpg

[*] Loading segmentation mask: resilience/spacenet_generalize_mask.png

[*] 1. Constructing Mathematical Graph from Mask (sknw)...

-> Identified 44 intersections (Nodes)

-> Identified 42 road segments (Edges)

[*] 2. Calculating Structural Chokepoints (Betweenness Centrality)...

[*] 3. Running Targeted Attack (Cascading Overload)...

- Cascade Depth: 1 chain-reaction steps

- Total Roads Failed: 6

- Network Fragmentation: 50.0% capacity lost

[*] 4. Projecting Graph to Earth Coordinates...

[*] Projecting pixel graph onto Earth coordinates using default Mumbai bounding box...

[*] 5. Exporting GeoJSON for Kepler.gl: resilience/Network_Resilience_3D_MS_chipcustom.geojson

[*] GeoJSON successfully exported to: resilience/Network_Resilience_3D_MS_chipcustom.geojson

[*] 6. Generating 2D Geospatial Visualization Layer...

[*] 2D Visualization saved to: resilience/Network_Resilience_Analysis_MS_chipcustom.png

[+] Graph Resilience Analysis & Geo Export Complete!

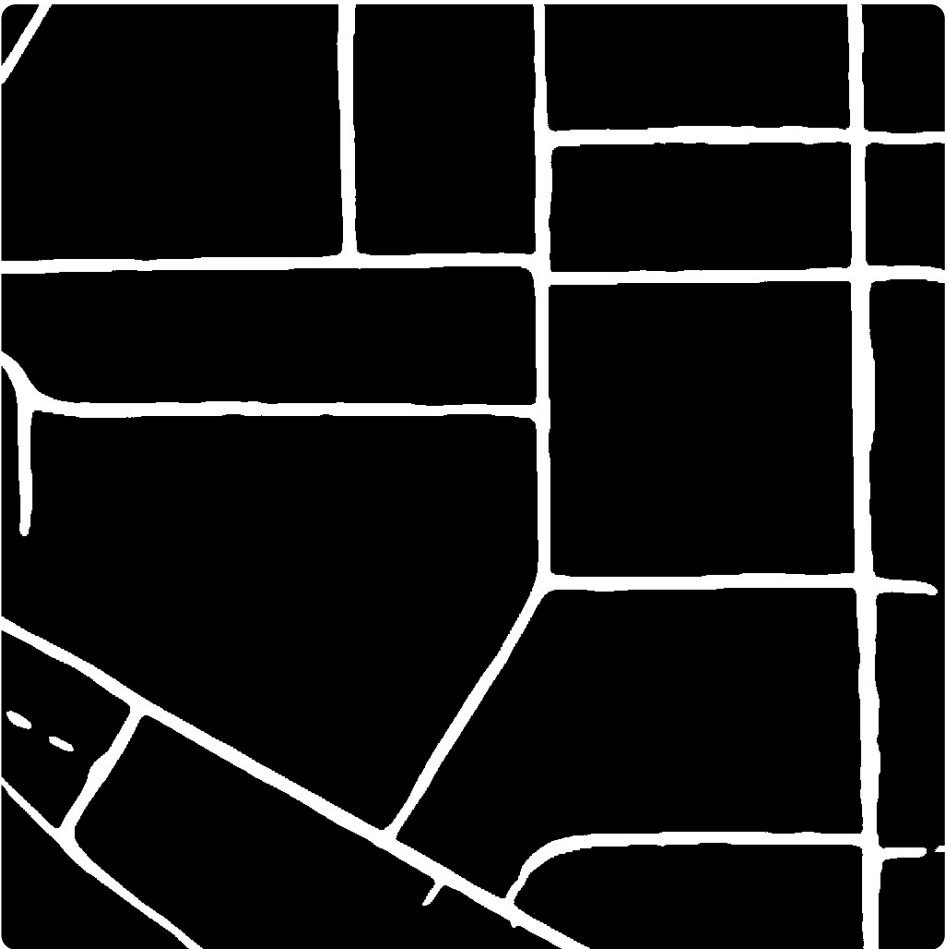
Pipeline Execution Complete!

Step-by-Step Visualizations

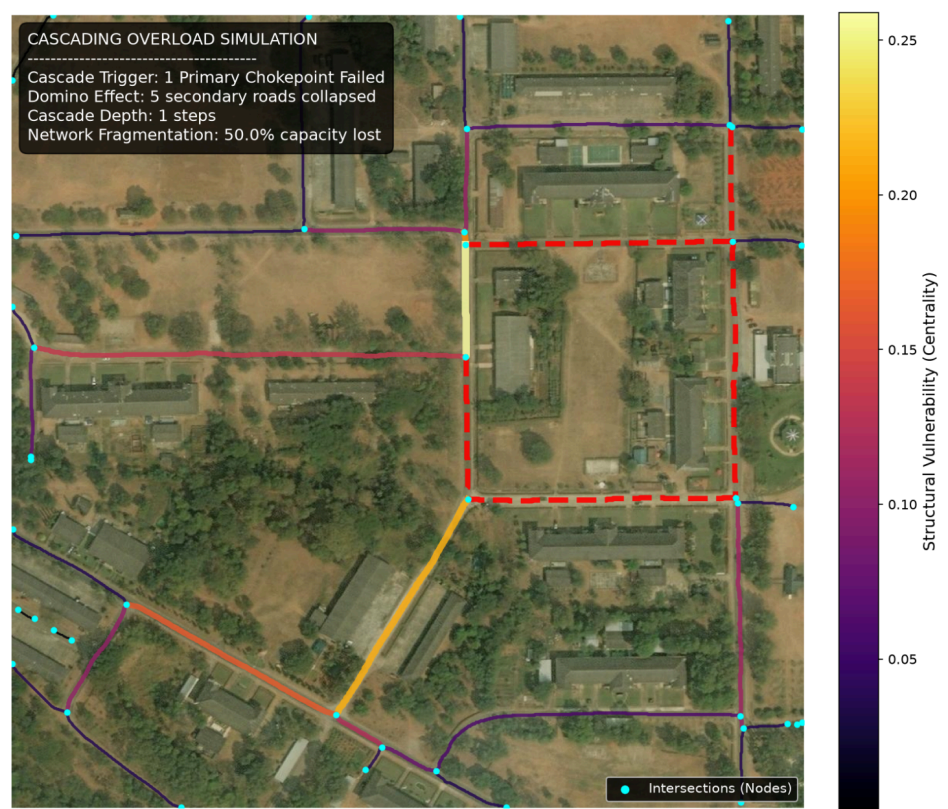
1. Input Satellite Image



2. AI Extracted Mask (UNet++)



3. Graph Analysis & Vulnerabilities



4. Threats & Results Summary

Simulation: Targeted (Choke Points)

Total Network Capacity Lost

50.0%

Cascading Steps

1

Roads Destroyed

6

Interactive 3D Network Map

Explore the extracted graph nodes and choke points projected onto Earth coordinates.

